

Program Structure and Output

L04 - Output

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- C++ uses **streams** to perform input and output operations.
- A stream is an entity where a C++ program can either insert or extract characters to or from.
- In this module, we will use the **cin** (to handle input) and the **cout** (to handle output) streams.

- In C++, output is handled using the **iostream** library.
 - This library should be included as a preprocessor directive whenever you want your program to be able to handle input or output.
- The library provides the definitions for the << operator as well definitions for the cout and endl operations.

- The operator `<<` is also known as the **insertion operator**.
- It indicates that whatever follows it is to be sent (piped) to the defined output channel.
- The output channel in this case is the console, hence the use of **`cout`** before the operator.

- In C++ **cout** is used together with the insertion operator << as shown below:

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     cout << "Welcome to COS-132" ;
5     return 0;
6 }
```

- Notice that the statement is enclosed in double quotes because it is a "string literal"

```
1 #include <iostream>
2
3 using namespace std;
4
5 int main() {
6     cout << 123;
7     return 0;
8 }
```

- Numbers

- In C++ you may use multiple insertion operators in a single statement as shown below:

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     cout << "COS-132"<< " - Module" ;
5     return 0;
6 }
```

- When the above program is run, what will be the output?

- To insert a line break, you may use the new line character: .

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     cout << "COS-132-\n";
5     cout << "Module";
6     return 0;
7 }
```

- You need to type it at the exact position the line must be broken.

- As an alternative option you may use the endl manipulator.

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     cout << "COS-132" << endl;
5     cout << "Module";
6     return 0;
7 }
```

Study the following code example. Will the following program compile? If not, what seems to be the problem?

```
1 #include <iostream>
2
3 using namespace std;
4
5 int main() {
6     cout >> "COS132-2024" ;
7     return 0;
8 }
```

Compile and run the program below:

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     cout << "Hello" << endl ;
5     cout << endl ;
6     cout << "Joan" << endl ;
7     return 0;
8 }
```

- Now remove the `cout << endl;` statement and run the program again. How does the output change?